

Experiment name:Implementation of a simple http server to response html webpage

Theory:An HTTP server is a software application that listens for client requests from a web browser and sends responses using the HTTP (HyperText Transfer Protocol). A simple HTTP server can respond with an HTML webpage when a client accesses it through a browser.The purpose of implementing a simple HTTP server is to understand the basic working mechanism of web communication between a client and a server

Code:

```
import java.io.BufferedReader;
import java.io.IOException;
import java.io.InputStreamReader;
import java.io.OutputStream;
import java.net.ServerSocket;
import java.net.Socket;
import java.nio.charset.StandardCharsets;

public class SimpleHttpServer {
    private static final int PORT = 8080;

    public static void main(String[] args) {
        System.out.println("Simple HTTP server started on
http://localhost:" + PORT);

        try (ServerSocket serverSocket = new ServerSocket(PORT)) {
            while (true) {
                Socket clientSocket = serverSocket.accept();
                handleClient(clientSocket);
            }
        } catch (IOException e) {
            System.err.println("Server error: " + e.getMessage());
        }
    }

    private static void handleClient(Socket clientSocket) {
        try (Socket socket = clientSocket;
```

```

        BufferedReader reader = new BufferedReader(new
InputStreamReader(socket.getInputStream(), StandardCharsets.UTF_8));
        OutputStream outputStream = socket.getOutputStream() {

String requestLine = reader.readLine();
if (requestLine == null || requestLine.isBlank()) {
    return;
}

while (reader.ready()) {
    String headerLine = reader.readLine();
    if (headerLine == null || headerLine.isEmpty()) {
        break;
    }
}

String html = "<!DOCTYPE html>"
    + "<html lang=\"en\">"
    + "<head>"
    + "<meta charset=\"UTF-8\">"
    + "<meta name=\"viewport\"
content=\"width=device-width, initial-scale=1.0\">"
    + "<title>Simple HTTP Server</title>"
    + "<style>"
    +
"body{font-family:Arial,sans-serif;background:#f4f7fb;margin:0;display:fl
e
x;min-height:100vh;align-items:center;justify-content:center;color:#1f2937
;}"
    + ".card{background:#fff;padding:32px
40px;border-radius:16px;box-shadow:0 10px 30px
rgba(0,0,0,.12);text-align:center;max-width:600px;}"
    + "h1{margin-top:0;color:#0f766e;}"
    + "p{line-height:1.6;font-size:1.05rem;}"
    + "</style>"
    + "</head>"
    + "<body>"
    + "<div class=\"card\">"
    + "<h1>It works!</h1>"
    + "<p>This HTML page is being served by a simple Java
HTTP server running on port 8080.</p>"

```

```

        + "<p>ID: 223311246</p>"
        + "</div>"
        + "</body>"
        + "</html>";

byte[] body = html.getBytes(StandardCharsets.UTF_8);
String response = "HTTP/1.1 200 OK\r\n"
    + "Content-Type: text/html; charset=UTF-8\r\n"
    + "Content-Length: " + body.length + "\r\n"
    + "Connection: close\r\n"
    + "\r\n";

outputStream.write(response.getBytes(StandardCharsets.UTF_8));
outputStream.write(body);
outputStream.flush();
} catch (IOException e) {
    System.err.println("Client error: " + e.getMessage());
}
}
}

```

Output:

```

shuvo@shuvo-HP-EliteBook-840-G3:~/Documents/parra processing la
o b$ cd "/home/shuvo/Documents/parra processing lab/simple http s
  erver to response html webpage/" && javac SimpleHttpServer.java
  && java SimpleHttpServer
  Simple HTTP server started on http://localhost:8080
  □

```

It works!

This HTML page is being served by a simple Java HTTP server running on port 8080.

ID: 223311246